RISHI DINESHCHANDRA SALAT

AUTONOMOUS DRIVING & MECHANICAL ENGINEERING

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Portfolio – projects, results & certificates: rishisalat.github.io/rishisalat

PROFILE

Mechanical Engineering graduate who moved into Autonomous Driving out of a genuine interest in AI and ADAS – technologies that are rapidly reshaping the vehicle industry. With hands-on experience in vehicle diagnostics, CNC machine assembly, and systems integration, I bring a practical engineering mindset to autonomous systems. Currently pursuing a Master's in Autonomous Driving at Hochschule Coburg, actively building ROS2-based robotics systems, automated test pipelines, and object detection on NVIDIA Jetson. Looking for a Werkstudent or internship role on the hardware and systems side of autonomous vehicles.

AREAS OF EXPERTISE

VEHICLE & MECHANICAL SYSTEMS	Vehicle Diagnostics | Vehicle Maintenance | Engine & Transmission Basics | Vehicle Inspection | Troubleshooting | Workshop Operations | Safety Compliance
AUTONOMOUS DRIVING & ROBOTICS	ROS2 | Robotics | Localization | Object Detection (SSD-MobileNet) | NVIDIA Jetson | Autonomous Driving Systems | LiDAR | OptiTrack
AUTOMATION & MANUFACTURING	CNC Machine Assembly | Machine Integration | Mechanical & Electrical Assembly | Functional Testing | Commissioning | Quality Control | Manufacturing Processes | Precision Measurement
SOFTWARE & TOOLS	Python (Basic) | Git / GitLab CI | pytest | Roboflow | AutoCAD | MS Office (Excel, Word, PowerPoint) | CRM Software | Vibe Coding

EXPERIENCE

Technician, *05/2025 – 10/2025*

Saurashtra Motors – Vadodara, India

- Diagnosed and repaired petrol and diesel vehicles, covering engine, transmission, and mechanical faults.
- Carried out routine inspections and caught mechanical issues early, reducing repeat service visits.
- Handled direct communication with customers on fault findings, repair timelines, and service recommendations.
- Maintained workshop safety and compliance standards throughout all service operations.

Trainee Engineer, *01/2025 – 04/2025*

Solar CNC Automation – Vadodara, India

- Assembled 2–3 CNC machines end-to-end, handling both mechanical structure and electrical wiring.
- Led functional testing and commissioning of a 5-axis CNC machine, verifying all axes moved to specification.
- Performed alignment checks, cable routing, and control system integration during machine build.
- Identified and resolved assembly defects during quality checks before final handover.

Intern, *06/2024 – 07/2024*

Vivek Steel Traders – India

- Operated conventional lathe machines for metal component production.
- Used Vernier calipers and micrometers for precision measurement and quality verification.
- First hands-on exposure to manufacturing floor operations and workshop safety protocols.

EDUCATION

Master of Engineering – Autonomous Driving *03/2026 – Present*

Hochschule Coburg – Kronach, Germany

- Full-time studies

Bachelor of Engineering – Mechanical Engineering *09/2021 – 05/2025*

Gujarat Technological University – Vadodara, India

Neotech Technical Campus

- CGPA: 8.3 / 10 | German Grade equivalent: 1.8

PROJECTS

AEGIS Robots – Autonomous Indoor Robot, *Ongoing*

Hochschule Coburg – Master's Project | *Code & write-up: github.com/rishisalat*

- Established the project identity – created the AEGIS name, logo, and prototype visuals used in team presentations.
- Independently wrote the publisher/subscriber node for the robot's localization component in ROS2, providing the robot pose to the planning components.
- Built the GitLab CI pipeline for the localization component: pre-commit linting, automated ROS2 node tests, and an 80% test coverage gate.
- Building a 1,200-image dataset of 5 indoor object classes (table, chair, person, dustbin, bottle), annotated in Roboflow, and training an SSD-MobileNet detector on the robot's NVIDIA Jetson.
- Integrated LiDAR and OptiTrack positioning systems to enable indoor autonomous navigation.

Transfer Learning – Object Detection on a Model Car, *10/2026*

Hochschule Coburg – Advanced AI Course (team of 2)

- Collected and annotated an own dataset in Roboflow; fine-tuned SSD-MobileNet on a model car's Jetson.
- Analysed why the assigned training parameters made the model forget its pretrained knowledge (mAP 0.597 → 0.007) and proved it with a control run that reached mAP 0.474.

Portfolio Website & GitHub Profile, *10/2026*

- Built my portfolio website and GitHub profile through vibe coding: planned the content, directed the design and published it on GitHub Pages.

Hybrid Energy Generator, *Undergraduate*

- Designed a conceptual hybrid mechanical-electrical energy system as part of coursework.
- Carried out efficiency analysis and feasibility evaluation of the proposed concept.

Pollution Control in Commercial Vehicles, *Undergraduate*

- Developed an exhaust treatment concept to reduce emissions in commercial vehicles.
- Researched filtration methods and built a preliminary prototype concept with feasibility assessment.

Marketing Project,

Neotech Technical Campus – Vadodara, India

- Ran student recruitment outreach, managing communication workflows and data reporting.

LANGUAGES

- English
- Hindi
- Gujarati
- German – A1 (Learning)